Mid Quarter project progress Report (Group 5)

Dataset Description: Our team uses the IMDB noncommercial database to create a machine learning model that predicts a movie rating based on impactful features before release. To predict IMDb movie ratings, we initially selected features—actors, directors, writers, budget, language, genre, and runtime—based on their established influence on movie success and audience reception. Our choice of initial features was inspired by research and previous work on creating Machine Learning models predicting movie ratings, evident in [1]. Within the IMDB Database, the datasets available are 'title.akas.tsv.gz' and 'title.basics.tsv.gz', providing information about movie titles,'title.principals.tsv.gz', 'name.basics.tsv.gz', and 'title.crew.tsv.gz', providing information about actors, directors and writers, and 'title.ratings.tsv.gz' with information about the rating of a movie. The budget data was missing from the IMDB database. Because of the implied importance of budget in predicting the rating, we decided to use a movie budget dataset from Kaggle, which we could cross-reference with the IMDB database. Specifically, we were using the train.csv file's unique identifier 'imdb_id' to associate with the IMDB unique identifier 'Tconst', which allowed us to merge the two datasets. The primary objective during the Exploratory Data Analysis was to collect, analyze, and impute these features for training our model. Furthermore, we aimed to demonstrate a strong relationship between features and IMDb ratings, which was achieved with reliable data.

Exploratory Data Analysis and Data Cleaning: Exploratory Data Analysis and Data Cleaning were accomplished, so each team member was assigned one or two features for which they conducted detailed analysis. Based on the merged data, the initial features with a correlation lower than 0.05, including runtime and language, were removed because their explanatory value was minimal. The following is the analysis of the features that have been shown to be of high importance to our project.

Budget vs. IMDb Moving Rating: After to analysis examined the relationship between a movie's production budget and its IMDb user rating to evaluate whether financial investment impacts perceived movie quality. My initial dataset included 3,000 films (Budget dataset from Kaggle), but there has 811 of them reported a budget as $0, which is often indicative the budget value has missing or unreliable data. So, after I was removed those data, leaving 2,188 valid movies for analysis. Moreover, I'm trying to address the wide range value in budget that will cause strong right-skewness in the raw budget values. Thus, I'm using the base-10 logarithmic transformation the budget dataset vs rating. This transformation will normalize the distribution, reducing the influence of extreme values and resulting in a more symmetric spread. The post-transformation histogram showed that most log_10 budget (Fig6) values clustered between 6.0 and 8.0, which corresponds to real-world budgets ranging from $1 million to $100 million. Despite this transformation, a few high-budget films remained as outliers (Fig5), which were clearly visible in the boxplot and could potentially bias model training if not handled appropriately. Hence, based on the graphic of the relationship, I'm really confidence to say that the budge has the correlation with rating.

Actor Score vs. IMDB Movie Rating: This section of EDA and Feature Selection quantified the impact of actors on movie ratings using the IMDb non-commercial database. The core of the analysis is the actor scores table, which captures actors' performance and experience through features like average movie rating, average votes per movie, total movies, movie rating standard deviation, high-rated movie count, career span, and recent average rating (last 5 years). A correlation heatmap in Figure 1 of these features shown that there is a high correlation between the average movie rating and recent average movie ratings, suggesting future models should use only one of the features.

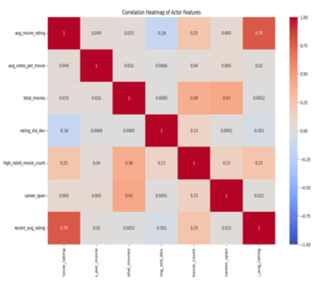


Fig 1

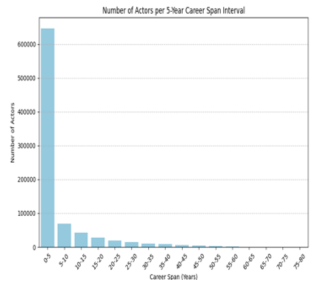
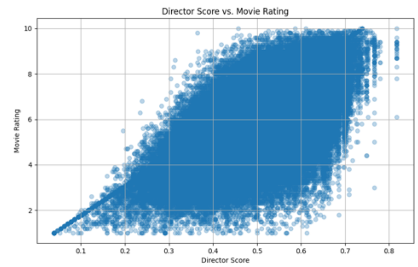


fig2



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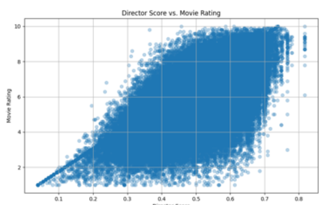

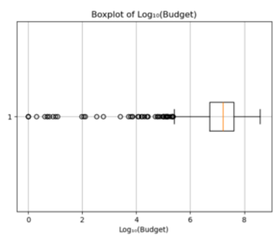

Fig4 fig5 fig6

Data preprocessing addressed quality issues in the dataset. Actors with fewer than three movies (650,000 with one movie, 110,000 with two) were excluded to reduce noise, as their limited features skewed the data, confirmed by right-skewed distributions in plots of actors per 5-year career span and total movies in Figure 2. Of the remaining 222,531 actors, 17,199 actors had missing rating standard deviation and were imputed using the average rating standard deviation for actors with a 0–5 year career span. Additionally, 142,180 actors lacked a recent average rating, and were imputed with the average across all actors. Actors with career spans over 80 years were removed to eliminate data errors.

The cleaned dataset was used to train a Ridge regression model on movies from 2020 to 2025, mitigating the risk of inflated actor scores for older movies (e.g., from 10 years ago) due to using all historical data. Analysis of the IMDB title principals table confirmed that higher ordering (1–2) corresponds to leading actors, while lower ordering (3–7) indicates supporting cast, validating the aggregation of scores for the top 7 actors. The average actor score of these actors formed each movie's average actor score feature, representing cast quality. The Ridge regression model achieved an MSE of 0.6638, RMSE of 0.8148, and R^2 of 0.7283, explaining 72.83% of the variance in movie ratings. The model's weights were used to compute each actor's final actor score as a weighted sum of standardized features, scaled and shifted to ensure non-negative values, facilitating interpretable predictions of movie ratings. In the future, the actor scores table will be created using a time-aware approach, where for each movie, actor features will contain information about the actor until the movie is released. Aggregate actor scores per movie will be used to train the final model.

Movie Director Score vs. IMDB Movie Rating: The analysis of the movie director's impact on movie ratings was investigated through director features, quantifying a director's performance and experience. These features include: average rating of a director's movies, total number of movies directed, and aggregate number of votes across all movies. The analysis revealed 4,094 movies with missing director information, which were excluded as they couldn't contribute to the director analysis. Additionally, 33 movies had missing release years, which were imputed with the median year (2007). After data cleaning, the analysis processed 371,406 director-movie pairs across 146,578 unique directors. For each director, a comprehensive score was calculated using a weighted formula that balanced their popularity and experience, with 70% weighted on average rating and 30% on movie count. The resulting director scores showed a strong correlation with movie ratings, with an R2 value of 0.435, demonstrating that the director substantially impacts a film's quality. To quantify the impact of director scores on movie ratings, we built a Ridge regression model to predict movie ratings based solely on director scores exclusively. The model achieved Mean Squared Error (MSE) of 1.0751, Root Mean Squared Error (RMSE) of 1.0369 and R2 of 0.4351.

Genre vs. IDMB Movie Rating: My analysis investigates how IMDb rating is influenced by film genre and the results appear as follows: Inside each genre the middle score (median) may demonstrate distinct differences from one another. The rating scale for documentaries reaches approximately 7.3 whereas horror movies generally stay at 4.9. About 2.4 points exist between the highest rated film genre which gets a 7.3 compared to the lowest rated genre which gets a 4.9 according to the 1–10 scale. Most interesting genres receive a rating between 5.6 and 6.5 on IMDb rating systems. I also discovered that the score range in which half of all ratings are situated within each genre extends from 1.0 to 1.5 points. Ratings are affected equally by both the choice of director and actors together with script quality and specific genres. Every other category maintains ratings that lie approximately half a point from the average level which makes them sit in the middle range. The ratings range of different genres can be explained by this single category to the extent of 10–20%. The majority (80–90%) of rating differences emerge from elements beyond generic classification such as storytelling and performance alongside directing and other factors.Wide Vertical Spread:Every type of media shows a complete distribution of ratings across all scales from one to ten.

Literature Review: Couple people have built imdb rating predictors before. We happened to have used a very similar approach as them to import and clean the data. The later stages of our project will differ a lot from the previous approaches since we are planning on training DNNs for our model. So far no one has used this approach with the previous projects either using logistic regression, k nearest neighbors algorithm or a tree algorithm like random forest. We will also try developing models for these algorithms but we suspect that a DNN might be better at predicting the ratings due to the complexity of all the variables involved (actor, director, etc). The projects before us concluded that tree algorithms like random forest and gradient boosting were the most accurate at predicting imdb ratings with an accuracy of about 75%. Even though we haven't covered those techniques in class we can maybe also try building these models to compare with our other models. Overall our goal is to reach a higher accuracy of at least 80 - 90% to improve on the previous projects. We should be able to achieve this with the use of DNNs.

Team distribution report: Tianren Chen contribute with dataset description and investigated Budget and Runtim, Sasa Antunovic contribute with EDA and Actor contibution, Yanming Luo contribute with feature selection and Genre, Language , Navroop Bath contribute with literature review and Writers feature. Finally, Safwan Ali contribute as 0%, no participate in report, contributed with Directors feature.

Reference

[1] W. R. Bristi, Z. Zaman, and N. Sultana, “Predicting imdb rating of movies by Machine Learning Techniques,” *2019 10th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, Jul. 2019. doi:10.1109/icccnt45670.2019.8944604

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